

⚡ PeopleSoft Trace Insights (Browser Edition)

Enterprise Trace Diagnostic Engine, Execution Path Diffing & RCA Guide

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[100% Client-Side In-Memory Engine] • [Zero Data Uploads] • [SQL, PPC & AE Analysis]

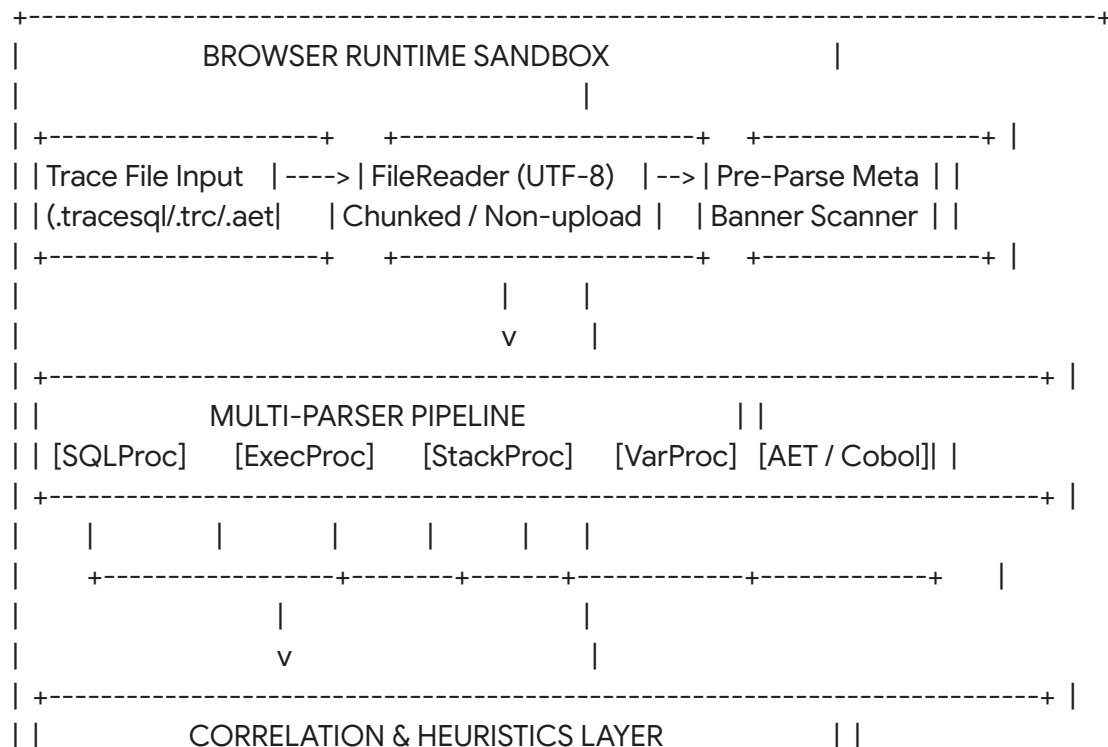
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1. Tool Overview & Architectural Concept

PeopleSoft Trace Insights (Browser Edition) is an enterprise diagnostic platform packaged into a self-contained, single-file HTML application. It operates **100% client-side** directly within the web browser using modern Web APIs (FileReader, TypedArray, Performance, Int32Array).



```
| | • Variable -> Bind Correlation • Cursor Lifecycle Engine | |
| | • Multi-RDBMS Error Classifier • Fingerprint & N+1 Pattern Analyzer | |
| | • PeopleCode Hotspot Profiler • Myers Path Alignment Diff Engine | |
| +-----+
|           |           |
|           v           |
| +-----+
| |           DYNAMIC PRESENTATION WORKSPACE           | |
| | [Overview & RCA] [Statistics] [SQL Grid] [Exec Tree] [Diagnostics] [Diff] | |
| +-----+
+-----+
```

2. Complete User Interface (UI) Architecture

The user interface uses a responsive layout with custom design tokens for both Dark and Light modes.

```
+-----+
| [ ⚡ Trace Insights] PeopleSoft Trace Insights [ 🏠 Home] [ ⚡ Compare] [ 💾 CSV] |
|           Browser Edition           [ 🌞 Light] [ ? Help] [mem: 42MB] |
+-----+
| [ 📊 Overview] [ 📈 Stats] [ 📄 SQL (1.2k)] [ 🌲 Exec Path] [ 🚨 Issues (4)] [ 📋 Vars] |
+-----+
| [All] [Slow] [Errors] [Repeated] [High Fetch] [By Where] [By From] [By Tables] |
+-----+
|           ACTIVE PANEL WORKSPACE           |
+-----+
| (c) 2026 PeopleSoft Trace Insights           Developer Contact: Sudheer Putti |
+-----+
```

2.1 Top Navigation Header (#hdr)

- **Branding Logo (.logo):** Displays ⚡ Trace Insights with an accent-to-purple gradient.
- **Subtitle (.subtitle):** Identifies the tool as PeopleSoft Trace Insights — Browser Edition.
- **Right Control Hub (#hdr-r):**
 - 🏠 Home (goHome()): Resets workspace, clears comparison states, and returns to the file ingestion landing view.
 - ⚡ Compare Traces (openCompareModal()): Opens the two-file path comparison modal.

-  CSV (exportCSV()): Exports parsed SQL telemetry and fingerprint diagnostics directly to CSV.
-  Light /  Dark (toggleTheme()): Toggles between the high-contrast slate dark theme (#0b1120) and clean light theme (#f8f9fc).
-  Help (openHelp()): Opens the modal containing PeopleTools trace bitmask references and troubleshooting recipes.
- mem: [X] MB (#mem): Real-time monitor displaying active V8 JavaScript heap usage via performance.memory.

2.2 Landing & Drag-and-Drop Zone (#dz)

Displayed on initial launch and when returning Home:

- **Drop Card (.dcard)**: Interactive target zone supporting native drag-and-drop events (dragover, dragleave, drop).
- **Folder Glyph (.di)**: Large highlighted icon () with drop-shadow glow.
- **Headline & Sub-description (.dt, .ds)**: Informs the user that files (.tracesql, .trc, .aet, .txt, .log) are executed 100% locally in client memory with zero server uploads.
- **Browse File Button (.btn-p)**: Launches the hidden native file input (#fi).
- **Progress Bar Wrapper (#pw)**:
 - **Status Label (#pl)**: Displays dynamic loading text (e.g., "Reading [file]...", "Parsing trace execution...").
 - **Track & Fill Bar (#pbo, #pbi)**: Visual feedback bar scaling from 0% to 100%.

2.3 Primary Workspace Navigation Tabs (#tbar)

Fixed horizontal tab strip for toggling analytical panels:

1.  **Overview (#t-overview)**: High-level telemetry, automated RCA, trace findings, and system profiles.
2.  **Statistics (#t-stats)**: Aggregated metrics, database operation ratios, and execution distribution charts.
3.  **SQL Statements (#t-sql)**: Comprehensive tabular statement grid with counter badge (#bg-sql).
4.  **Execution Path (#t-exec)**: Hierarchical call tree mapping PeopleCode, App Engine actions, and SQL statements.
5.  **Diagnostics & Issues (#t-stack)**: Captured exception stack traces, buffer discrepancies, and value issues with badge (#bg-st).
6.  **Variable Flow (#t-vars)**: Timeline tracking PeopleCode variables, assignments, and buffer field fetches with badge (#bg-var).

2.4 SQL Filter Bar & Subtabs (#stabs)

Rendered automatically when the **SQL Statements** tab is active:

- All (#s-all): Displays every parsed statement.
- Slow (#s-slow): Statements with execution duration $\geq 1.0\text{ s}$.

- High Fetch Calls (#s-fetch): Queries returning high volume (fetch count ≥ 100).
- By Where (#s-where): Statements grouped by WHERE clause syntax.
- By From (#s-from): Statements grouped by target table/view FROM expressions.
- Tables (By Type) (#s-tbl): Categorized matrix of tables operated on by SELECT, INSERT, UPDATE, and DELETE.

2.5 Global Search Filter Bar (#sb)

- **Search Input (#si):** Real-time substring filter triggered across visible table rows or by pressing Ctrl+F (Cmd+F on macOS).
- **Match Indicator (#sc):** Live counter showing matching row counts.
- **Close Button (#sclose):** Clears search inputs and restores full row visibility.

2.6 Executive Overview & RCA Dashboard (#p-overview)

```

+-----+
| [ Trace Health: 85/100 ] [ SQL Time: 12.450s ] [ Slow: 3 ] [ Diagnostics: 2 ] |
+-----+
| 🧠 Root Cause Analysis & Remediation      [SQL ✓] [Binds ✓] [Timings ✓] [PPC ✓] |
+-----+
| 🔴 SQL_ERROR: Unique Constraint Violation (ORA-00001) |
| Symptom: INSERT INTO PS_EX_LINE failed at line 14205. |
| Fix: Check unique keys on PS_EX_LINE in Application Designer. |
| [View trace evidence] |
+-----+
| 🔍 Trace Findings (Prioritized) | 📄 Trace Profile & Parameters |
| • Database Errors Detected (1 statement) | • File Name: hr_hire_online.tracesql |
| • Potential Event Loop Detected (Save) | • Tools Release: 8.60.07 |
| • Repeated SQL Patterns (N+1 candidate) | • Database: HRPROD (ORACLE) |
| • TraceSQL: 3 (0x3) [Stamped] |
| • TracePC: 1 (0x1) [Inferred] |
+-----+

```

- **KPI Summary Grid (.dashboard-grid):**
 - Trace Health: Computed **0–100** score penalized by SQL errors, unhandled exceptions, slow queries, high fetch bursts, and recursive event loops.
 - SQL Execution Time: Total attributed database time and statement volume.
 - Slow Queries: Statements exceeding the **1.0 s** threshold.
 - Diagnostics & Errors: Total count of caught exceptions, failed SQL calls, and value

warnings.

- **Root Cause Analysis & Remediation Card (.rca-card):**
 - **Coverage Badges:** Visual pills verifying the presence of SQL, Binds, Timings, and PeopleCode data in the trace.
 - **Synthesized Findings:** Categorized issues (e.g., AE_ABNORMAL_TERMINATION, FATAL_EXCEPTION, SQL_ERROR, SQL_NPLUS1, TRACE_FLAG_GAP, UNCONSTRAINED_SQL, PEOPLECODE_HOTSPOT).
 - **Prescribed Remediations:** Direct guidance and actionable recommendations.
 - **Trace Evidence Launcher:** Button opening raw file lines associated with each issue.
- **Trace Findings Card:** Severity-coded indicators (dot-critical, dot-warning, dot-info, dot-success).
- **Trace Profile & Parameters Card:** Metadata panel displaying PeopleTools release, Database Name/RDBMS type, Operator ID, Batch/Online classification, total lines, duration metric (*s* or *ms*), and explicit/inferred bitmask flags (TraceSQL, TracePC, TraceAE).
- **Top SQL Hotspots Card:** Statements ranked by total consumed time.

2.7 Statistics & Visual Offline Metric Charts (#p-stats)

- **Metric Cards Grid (#sc-stats):** Groups metrics into categories:
 - SQL Statements: Counts, aggregate durations, longest single statement, maximum row fetches, error counts.
 - SQL by Type: Breakdown by SELECT, UPDATE, INSERT, DELETE.
 - Execution Path: Maximum nesting depth, total PeopleCode calls, exceptions count, buffer deserializations.
 - Transaction Control: Commits, Rollbacks, Connects, Disconnects.
 - PeopleCode Hotspots: Top event methods ranked by inclusive runtime.
 - Cursor Lifecycle: Active database cursor handles, fetch distributions, and errors.
 - Temporary Tables: Heuristically identified temp tables (_TAO, _TMP, _TEMP).
- **Visual Charts Container (.chart-container):**
 - **Time Distribution Chart (#chartDist):** Visual distribution bar comparing Database SQL Time vs. Application PeopleCode Time.
 - **Top 10 Slowest Operations Chart (#chartTop):** Ranked bar chart highlighting the top 10 bottlenecks with direct links to call lines.

2.8 Interactive SQL Data Grid (#p-sql)

Dynamic data table featuring sortable column headers, text ellipsis clipping, and execution time distribution bars:

- **Standard Columns:** Line Number, Duration (with proportional color-graded progress bars: green → yellow → red), SQL % of total time, Fetch Count, Operation Type, SQL Statement Text.
- **Fingerprints / Repeated View:** Executions, Total Time, SQL %, Average / P95 / P99 Latency, Max Single Duration, Normalized Pattern.

- **Where / From Grouping:** Aggregate execution count, total duration, and clause text.
- **Tables View:** Operation Type, Referenced Tables, Total Query Count.

2.9 Execution Call Tree Workspace (#p-exec)

- **Execution Toolbar (#exec-toolbar):** Displays execution telemetry (Contexts, Nodes, Max Depth, Exceptions) alongside Expand All and Collapse All tree controls.
- **Hierarchical Tree Container (#epc):**
 - Indented tree structure using CSS nested margin layouts.
 - **Type Badges:** PC (PeopleCode Event), EXT (External Function), CALL (Internal Routine), SQL (Correlated Database Query).
 - **Color Highlights:** Red for error nodes/parents of errors (.ne), Yellow for slow executions ≥ 1.0 s (.nh), and Normal (.ns).
 - **Inline Code Preview:** Auto-extracts captured PeopleCode statements mapped directly under their execution nodes.

2.10 Diagnostics, Stack Traces & Value Issues Panel (#p-stack)

- **Stack Trace Cards (.ste):** Red border cards showing caught exceptions, offending objects, trace line locations, and full call frame listings.
- **Value Diagnostic Cards (.ste.warn):** Yellow border warning cards flagging uninitialized variables, null buffer field fetches, empty fetches, null binds, and numeric precision overflows.
- **Buffer Snapshot Cards:** Displays captured serialized/deserialized component and global buffer hierarchies.

2.11 Variable & Field Timeline Flow Panel (#p-vars)

- **Variable Search Bar (.var-input):** Autocomplete input supported by an HTML5 <datalist> containing captured variable and field names.
- **Timeline Container (#varTimelineContainer):** Chronological sequence cards displaying:
 - Action Type: FETCH, STORE, ASSIGN (&var = val), FIELD_ASSIGN (RECORD.FIELD = val).
 - Variable Identifier & Line Number.
 - Serialized Value string.
 - Direct deep link button: View correlated SQL (connects buffer values directly to subsequent SQL binds).

2.12 Execution Path Comparison Page & Modal (#comparePage, #compareModal)

```
+-----+
|  Compare Paths Execution-path comparison & divergence [Change Files] [Back] |
+-----+
| Trace 1: Base.tracesql (450 steps) | Trace 2: Target.tracesql (472 steps) | Dev: 22 |
+-----+
```

⚠ Execution Path Deviation Identified at Step #84		
Last Synchronous Step: PPC DERIVED_HR.CALC_BTN.FieldChange		
Probable Cause: Variable assignment or buffer field state differed.		
State & Field Variable Differences:		
Trace 1: &ALLOW_OVERRIDE = N (ASSIGN @ line 1240)		
Trace 2: &ALLOW_OVERRIDE = Y (ASSIGN @ line 1245)		
+-----+		
#	Trace 1 (Baseline)	Trace 2 (Target)
+-----+		
83	[MATCH] SQL SELECT [PS_JOB]	[MATCH] SQL SELECT [PS_JOB]
84	[REMOVED] PPC PROCESS_OVERRIDE	[ADDED] PPC LOG_BYPASS_AUDIT
(Click # to inspect discrepancy) (Click # to inspect discrepancy)		
+-----+		

- **Compare Modal (#compareModal):** Dual drop target enabling parallel file selection for Baseline (Trace 1) and Target (Trace 2).
- **Comparison Workspace (#comparePage):** Dedicated view isolating the comparative diff layout.
- **Divergence Summary Card (.diff-summary-card):**
 - Highlights the **exact step index** where execution logic branched.
 - Names the **Last Synchronous Step**.
 - Identifies the **Probable Cause** (e.g., bind mismatch vs. variable state mutation).
 - Displays side-by-side differences for bind values and variables.
- **Side-by-Side Diff Table (.diff-grid):**
 - Color-coded rows: MATCH (transparent), DIVERGE / ADDED (green tint), REMOVED (yellow/red tint).
 - Clickable Step Index (.diff-step-link): Opens the **Compare Step Difference Modal** for drill-down inspection of SQL, binds, variables, and code at that step.

2.13 Universal Slide-Out Detail Drawer (#drawer)

Fixed side panel (^{720 px} max width) opening over a backdrop:

- **Header:** Context-sensitive title and close button (×).
- **Content Formatter:**
 - Formatted SQL text with bind variables substituted back into the query.
 - Syntax highlighting: SQL Keywords (Teal), String literals (Yellow), Numbers (Orange).
 - Quick action: Copy SQL button.
 - Sub-sections: Binds, Table Lists, WHERE/FROM clauses, Stack Frames, Variable Correlation mappings.

2.14 Built-In Diagnostic Quick Reference Modal (#helpModal)

Contains PeopleTools trace configuration references:

- **Recommended Trace Recipes:** Reference table covering Online performance, AE control flow, IB transforms, and variable tracking.
- **Bitmask Matrices:** Detailed flag breakdowns for TraceSQL, TracePC, and TraceAE.
- **Diagnostic Principles:** Heuristic confidence rules and guidance on avoiding full scans.

2.15 Application Footer (#homeFooter)

Minimal persistent footer visible on the landing page:

- Copyright declaration (© 2026 PeopleSoft Trace Insights).
- Developer contact button linking directly to LinkedIn (Sudheer Putti).

3. How to Use the Tool

STEP 1: INGESTION	STEP 2: TRIAGE	STEP 3: INVESTIGATION	STEP 4: ACTION
+-----+	+-----+	+-----+	+-----+
Drop .tracesql, --> Review Health Score -> Analyze SQL Hotspots, -> Copy SQL, Fix	.trc, or .aet file & Automated RCA Call Tree & Variables Code & Export		
+-----+	+-----+	+-----+	+-----+

3.1 Phase 1: Loading & Ingesting Traces

1. Open the application in any modern desktop web browser (Chrome, Edge, Firefox, Safari).
2. On the main landing screen, drag your trace file into the dashed box, or click **Browse File...**
3. The real-time progress bar will indicate reading and parsing stages. Once complete, the workspace automatically opens.

3.2 Phase 2: Reviewing System Profile & Root Cause Analysis

1. Upon loading, review the  **Overview** tab.
2. Check the **Trace Health** score (out of 100):
 - **80–100** : Healthy run; focus on query optimizations.
 - **55–79** : Warnings present (slow queries, N+1 patterns, unconstrained WHERE clauses).
 - **0–54** : Critical conditions found (crashed Application Engine, unhandled exceptions, database errors).
3. Review the  **Root Cause Analysis & Remediation** card at the top:
 - Read the **Symptom** and **Actionable Fix**.
 - Click **View trace evidence** to open raw trace lines in the slide-out drawer.
4. Check the **Trace Profile & Parameters** card to confirm the PeopleTools Release, Database

platform, and whether trace flags were stamped or inferred.

3.3 Phase 3: Deep-Diving SQL Bottlenecks & N+1 Loops

1. Click the  **SQL Statements** tab.
2. Use the subtabs to filter statement types:
 - Click **Slow** to inspect statements taking longer than **1.0 s**.
 - Click **Repeated Patterns** to inspect normalized queries executed repeatedly inside loops. Look at the **N+1 Confidence** score and **Distinct Binds** count.
 - Click **Errors** to isolate failed SQL statements.
3. Use the search input or press Ctrl+F to search by table name (e.g., PS_JOB) or field name.
4. Click any statement row to open the **Detail Drawer**:
 - The drawer displays the complete query with **bind values substituted into place**, making it ready to run in SQL Developer, TOAD, or DBeaver.
 - Click **Copy SQL** to copy the query with bind values included.

3.4 Phase 4: Navigating the Execution Path Call Tree

1. Click the  **Execution Path** tab.
2. View the full hierarchical execution sequence across PeopleCode events, methods, and SQL calls.
3. Locate red-highlighted nodes to find where errors or exceptions occurred.
4. Use **Expand All** or **Collapse All** to navigate deeply nested routines.
5. Click any node to view its start/stop lines, inclusive runtime, and any captured PeopleCode source statements.

3.5 Phase 5: Diagnosing Stack Traces & Value Anomalies

1. Navigate to the  **Diagnostics & Issues** tab.
2. Inspect cards categorized under:
 - **Stack Traces**: Review unhandled PeopleCode exceptions and click to inspect the call stack down to the offending line.
 - **Value Issues**: Review uninitialized variables, null fetches, empty buffers, and numeric overflows that may cause logical branching errors.
 - **Buffer Snapshots**: Drill into serialized component buffer state dumps.

3.6 Phase 6: Tracing Variables & Field State Timeline

1. Switch to the  **Variable Flow** tab.
2. In the top search box, type a variable or field name (e.g., &EMPLID, EMPLID, &EFFDT) or select from the autocomplete dropdown.
3. Click **Filter Timeline** to see the chronological lifecycle of the variable (FETCH → STORE → ASSIGN).
4. If a variable was passed to a database call, click **View correlated SQL** to navigate directly

to the target statement.

3.7 Phase 7: Running Comparative Execution Diffs

1. Click  **Compare Traces** in the top navigation bar.
2. Trace 1 is automatically populated with your active trace.
3. Click **Choose Trace 2...** to select a comparison trace (e.g., a baseline trace from a working environment).
4. Click **Run Path Compare**.
5. The workspace opens the **Compare Paths** page:
 - Review the **Divergence Card** to see the first point of divergence and the likely cause (bind differences vs. variable state divergence).
 - Use the **Deviations Only** toggle to hide matching rows and focus on additions, removals, and logic branches.
 - Click any highlighted step index (#) to open the **Step Difference Dialog** for a side-by-side comparison of SQL, binds, variables, and code.

3.8 Phase 8: Exporting Diagnostic Telemetry

1. Click  **CSV** in the top navigation bar.
2. The tool generates and downloads trace_insights_diagnostics.csv.
3. Open the file in Microsoft Excel, Power BI, or Google Sheets for team sharing, auditing, and further performance analysis.

4. Comprehensive Feature Catalog

CORE FEATURE MATRIX AT A GLANCE	
Feature Area	Capability Summary
Security	100% In-Browser execution; zero network egress
Trace Compatibility	.tracesql, .trc, .aet, COBOL timing files
Automated RCA	Synthetic multi-factor issue detection engine
SQL Optimization	Bind parameter substitution, N+1 detection
Diff Analysis	Myers execution path alignment & divergence
Variable Flow	Cross-line field & buffer assignment tracking
Database Support	Oracle, Microsoft SQL Server, IBM DB2

4.1 Security & Zero-Upload Execution Model

- **Client-Side Processing:** Operates entirely in client memory using HTML5 FileReader. No trace data, SQL queries, operator IDs, or bind parameters are transmitted over the network.

- **Safe Large-File Handling:** Built-in heap guard prompts the user for verification when opening files exceeding **200 MB**, preventing browser tab crashes.

4.2 Pre-Parse Metadata Extraction & Flag Inference

- **Banner Inspection:** Automatically scans the first 2,000 lines of trace files before running the parser to extract:
 - PeopleTools Release (e.g., 8.59, 8.60, 8.61, 8.63).
 - Database Name & RDBMS Platform (ORACLE, MICROSOFT, DB2).
 - Operator ID (OPRID).
 - Batch Program Name (Application Engine process).
- **Heuristic Flag Reconciliation:** If explicit flags were omitted, the engine infers active bitmasks based on trace content:
 - Detects TraceSQL bitmasks: 1 (Statements), 2 (Binds), 4 (Connect/Commit), 8 (Fetch).
 - Detects TracePC bitmasks: 1 (Calls/Exec), 4 (Assignments), 8 (Fetches).
 - Detects TraceAE bitmasks: 1 (Steps), 2 (SQL), 128 (Timings).

4.3 Multi-Engine Parser Pipeline

- **Online SQL & PeopleCode Parser (SQLProc, ExecProc):** Parses COM Stmt=, EXE, Fetch, Bind-N, EPO, and ERR markers. Matches cursor identifiers across calls.
- **Application Engine Trace Parser (AETSQLProc, AETExecProc):** Parses .aet files, linking steps (PROGRAM.SECTION.STEP.ACTION) directly to SQL blocks, statement timings, and temporary table allocations.
- **COBOL Trace Parser (CobolSQLProc):** Parses batch timing reports, mapping cursor IDs, statement times, and fetch return codes.
- **Buffer Deserialization Parser (VarProc):** Reconstructs hierarchical component and global variable dumps.

4.4 Automated Root Cause Analysis (RCA) & Prescriptions

The analytical engine identifies issues and provides concrete remediation advice:

- **AE Abnormal Termination:** Detects batch crashes, identifies the offending step, and flags temporary table capacity issues.
- **Fatal Exceptions:** Maps unhandled PeopleCode errors back to the offending code object.
- **SQL Constraint Failures:** Flags unique index violations (ORA-00001, SQL Server 2627, DB2 803) and targets specific table keys.
- **Trace Bitmask Deficiencies:** Warns when bind tracing is missing (TraceSQL=3 recommended).
- **Unconstrained Full Table Operations:** Identifies large queries lacking a WHERE clause.
- **Loop / Event Recursion Alerts:** Detects recursive FieldChange and SaveEdit triggers.

4.5 SQL Fingerprinting & Heuristic N+1 Detection

- **Pattern Normalization:** Strips literals, comments, and numeric IDs to group queries into

canonical fingerprints.

- **N+1 Scoring Formula:** Calculates an N+1 probability score based on:

$$\text{Score} = f(\text{Execution Count}) + f(\text{Distinct Binds}) + f(\text{Consecutive Runs}) + f(\text{Shared Parent})$$

Flags iterative queries with a confidence rating (HIGH, MEDIUM, LOW).

- **Latency Distribution:** Computes Average, Median (P_{50}), P_{95} , and P_{99} latency metrics for each fingerprint.

4.6 Multi-RDBMS Error Classification Engine

Translates numeric error codes and database engine messages into standardized diagnostic categories:

- **UNIQUE_CONSTRAINT:** Oracle ORA-00001, SQL Server Msg 2627/2601, DB2 SQL0803N.
- **NULL_INSERT:** Oracle ORA-01400, SQL Server Msg 515, DB2 SQL0407N.
- **OBJECT_NOT_FOUND:** Oracle ORA-00942, SQL Server Msg 208, DB2 SQL0204N.
- **INVALID_IDENTIFIER:** Oracle ORA-00904, SQL Server Msg 207, DB2 SQL0206N.
- **BIND_ERROR:** Oracle ORA-01008.
- **NUMERIC_PRECISION:** Oracle ORA-01438.
- **VALUE_TOO_LARGE:** Oracle ORA-12899.

4.7 Execution Divergence & Myers Diff Algorithm

- **Myers Diff Implementation:** Uses typed arrays (Int32Array) to align execution paths of up to 3,500 steps per trace.
- **Root-Cause Divergence Pinpointing:** Automatically finds the point of divergence between two traces, highlights the last matching step, and identifies whether variable changes or bind values caused the branch.
- **Drill-Down Modal:** Allows users to inspect SQL text, bind differences, variables, and code snippets at any step.

4.8 Variable-to-Bind Correlation Engine

- Automatically traces values from PeopleCode assignments (&EMPLID = "KU0001") or buffer fetches through to their use in SQL bind parameters (Bind-1 ... value=KU0001).
- Displays confidence ratings (HIGH, MEDIUM, LOW) and line distances between assignment and SQL execution.

4.9 Cursor Lifecycle & Temp Table Profiling

- **Cursor Lifecycle Analyzer:** Monitors individual cursor IDs (Cur#N) across parse, bind, execute, and fetch cycles to detect unclosed cursors or fetch leaks.
- **Temporary Table Profiler:** Uses naming heuristics (PS_..._TAO, _TMP, _TEMP) to track temporary table operations and detect contention in Application Engine runs.

4.10 UI Accessibility, Design Tokens & Memory Guard

- **Dynamic Design Tokens:** Supports Dark and Light themes with persistent state saved in localStorage.
- **Zero External Dependencies:** Self-contained file requires no external CDNs, style sheets, or JavaScript libraries.
- **Live Heap Monitor:** Tracks browser memory usage in real time via the navigation header.

5. Summary Cheat Sheet: Trace Bitmasks

Flag	Value	Recommended Use
TraceSQL=3	1 + 2	Standard SQL Performance: Traces SQL statements (1) and bind variables (2).
TraceSQL=7	1 + 2 + 4	Transaction Analysis: Adds database Connect, Disconnect, Commit, and Rollback tracking (4).
TraceSQL=15	1 + 2 + 4 + 8	Fetch Analysis: Adds individual row fetch event logging (8).
TracePC=1	1	PeopleCode Flow: Traces program execution, function starts, and nesting flow.
TracePC=13	1 + 4 + 8	Variable Tracking: Adds assignments (4) and buffer fetches (8).
TracePC=3597	1+4+8+512+1024+2048	Deep PeopleCode Analysis: Adds parameter in/out tracking (512+1024) and attachment statements (2048).
TraceAE=3	1 + 2	Application Engine Flow: Traces step sequence (1)

		and SQL execution (2).
TraceAE=131	1 + 2 + 128	AE Performance Bottlenecks: Adds statement-level timings to trace file (128).
TraceAE=387	1 + 2 + 128 + 256	Complete AE Diagnostics: Adds PeopleCode action method timing detail (256).